RUGGEDISE THE HARDWARE, Increase Its Life

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McLaren-Honda, manufacturer of premium electric racing cars, recently started using silicon-carbide (SiC) power devices in inverters and on-board chargers, which increased their efficiency by 1.7 per cent while also reducing the cooling system size by 30 per cent. Thus, a change in the power component not only made their inverter compact but also rugged for use in electric cars. Similarly, gallium-nitride (GaN) power devices increase the reliability of low-power products.

"Both SiC and GaN-based power-conversion devices are creating a difference and making the product rugged," says Niranjan G., general manager (business development & technical marketing), ROHM Semiconductor.

External and internal ruggedness in hardware

A hardware system needs to be electrically as well as physically rugged so that it can withstand harsh environmental conditions, electrical faults, component failures, mechanical faults, etc.

For instance, for LED luminaires, designers select LED devices based on the application requirements of lumens, life-time, intensity and colour temperature. Optical component selection is also based on the application. Power supply is designed keeping in mind the desired current, voltage, power input, total harmonic distortion (THD) and power factor. Mechanical part is based on the thermal dissipation of the luminaire as a whole.

Besides, designers follow various manufacturing standards that are important from ruggedness point of view. LM79 standard is followed for LED luminaire performance test, and LM80 for LED evaluation. IPXX standards are followed for protection of LED luminaires and power supply from environmental conditions such as air, water, dust, physical contact and accidental contacts. IEC, EC, UL and BEE standards pertain to the quality of a hardware product.

The hardware includes the PCB and components on the PCB. Be it an analogue solution, power solution, sensing solution or mobile solution, hardware must be well-built and rugged.

Analogue and power solutions

There have been industrial upgrades in the quality of the PCB, power components and their hardware applications at electrical or physical level.

PCB and improved substrate. The PCB is an essential element of a hardware as it provides the base or the platform to the circuitry. The PCB material is chosen based on the type of application. For instance, as a small power supply with less input and output current will dissipate very less heat, it can be built on an FR4-based PCB, which costs very less.

Hrishikesh Kamat, CEO of Shalaka Connected Devices, says, "Multilayer boards have been prepped up to 256 layers by tech giants."

As electronic devices shrink, PCBs are

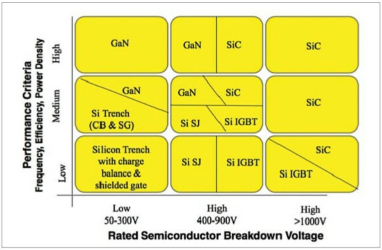

Fig. 1: SiC and GaN market analysis (Image courtesy: www.pntpower.com)